

Lecture 10

The Spectral Theorem

The sheet uses finite-dimensional real or complex inner-product spaces; quantum measurement questions are in the finite ideal-projective model. Inner products are in the physics convention; the adjoint satisfies $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$. A ★ marks a harder problem.

Core tutorial route

Work **10.4, 10.7, 10.8, 10.10, 10.16, 10.18, 10.22, and 10.25** in that order (about 80–100 minutes before discussion). This eight-problem route covers complex normal/Hermitian diagonalization, completion of the normal-triangular proof, the finite self-adjoint criterion, positive functional calculus, spectral projector interpolation, simultaneous commuting self-adjoint reasoning, a degenerate measurement transfer, and probability normalization.

Additional practice: 10.1, 10.3, 10.6, 10.9, 10.11–10.15, 10.17, 10.19–10.21, 10.23–10.24.

Bridge: 10.2 and 10.5 (real-entry normal operators viewed over $\mathbb{C}$; contrast their lack of real orthogonal diagonalization). **Extension:** none. Bridge tasks do not count toward core progress.

Diagonalization in an orthonormal basis

Exercise 10.1 (computational) Diagonalize the Hermitian matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ in an orthonormal basis: find its eigenvalues, an orthonormal eigenbasis, and the unitary U with $U^\dagger AU$ diagonal.

Exercise 10.2 (computational) On $\mathbb{C}^2$, show that $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ is normal but not self-adjoint, and find a unitary diagonalization. Explain why the same real matrix has no orthogonal diagonalization on $\mathbb{R}^2$.

Exercise 10.3 (computational) Diagonalize the real symmetric matrix $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ in an orthonormal basis: give the eigenvalues, an orthonormal eigenbasis, and the orthogonal O with $O^\top AO$ diagonal.

Exercise 10.4 (computational) Treat the following as a complex normal operator, unitarily diagonalize it, and then identify the stronger property that forces its eigenvalues to be real: $A = \begin{pmatrix} 1 & 2i \\ -2i & 1 \end{pmatrix}$: find its (real) eigenvalues, an orthonormal eigenbasis, and the unitary U with $U^\dagger AU$ diagonal.

Exercise 10.5 (computational) On $\mathbb{C}^2$, show that $A = \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$ is normal but not self-adjoint, and give a unitary diagonalization. Explain why there is no real orthogonal diagonalization on $\mathbb{R}^2$.

Exercise 10.6 (computational) Diagonalize the real symmetric $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}$ on $\mathbb{R}^3$ in an orthonormal basis, and give $O^\top AO$.

Exercise 10.7 (proof) In the proof of the finite complex spectral theorem, T is normal and upper-triangular in an orthonormal basis $e_1, \dots, e_n$. Show the first-row off-diagonal entries vanish, identify $A = a_{11} \oplus A_1$, prove the trailing block A_1 is normal, and complete the induction that A is diagonal.

Exercise 10.8 (proof) Let T be normal on a finite-dimensional complex inner-product space. Using the spectral theorem, prove that T is self-adjoint if and only if every eigenvalue of T is real.

Spectral decomposition

Exercise 10.9 (computational) Write the spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2$ of $\sigma_z = \text{diag}(1, -1)$, and verify $P_1 + P_2 = I$ and $P_1 P_2 = 0$.

Exercise 10.10 (computational) Using the spectral form, compute $f(A)$ for $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $f(\lambda) = \lambda^{1/2}$.

Exercise 10.11 (computational) Write the spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2$ of $A = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$, giving the projectors explicitly, and verify $P_1 + P_2 = I$ and $P_1 P_2 = 0$.

Exercise 10.12 (computational) Using the spectral form, compute the inverse A^{-1} of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Exercise 10.13 (computational) Find the positive square root $\sqrt{A}$ of the positive operator $A = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix}$ via the spectral form.

Exercise 10.14 (computational) Reconstruct the self-adjoint operator A on $\mathbb{C}^2$ with eigenvalue 3 on $\frac{1}{\sqrt{2}}(1, 1)$ and eigenvalue 7 on $\frac{1}{\sqrt{2}}(1, -1)$.

Exercise 10.15 (proof) Prove that an orthogonal projection P is self-adjoint and idempotent ($P^\dagger = P$, $P^2 = P$), and conversely.

Exercise 10.16 (proof) Let T be normal with distinct eigenvalues $\lambda_1, \dots, \lambda_m$ and spectral projections P_k . Prove the interpolation formula $P_k = \prod_{j \neq k} \frac{T - \lambda_j I}{\lambda_k - \lambda_j}$.

Measurement and commuting operators

Exercise 10.17 (computational) Measure σ_z on $|\psi\rangle = \frac{1}{\sqrt{10}}(|0\rangle + 3|1\rangle)$. Find the probabilities of ± 1 and the expectation $\langle \sigma_z \rangle$.

Exercise 10.18 (★) Let V be finite-dimensional. If $A, B \in \mathcal{L}(V)$ are commuting self-adjoint operators, prove that they have a common orthonormal eigenbasis.

Exercise 10.19 (computational) On $\mathbb{C}^3$, let $A = \text{diag}(1, 1, 4)$. Find the orthogonal projection onto the 1-eigenspace and the probability that measuring A on $|\psi\rangle = \frac{1}{\sqrt{3}}(1, 1, 1)$ yields 1.

Exercise 10.20 (computational) Measure σ_z on $|\psi\rangle = \frac{1}{2}(\sqrt{3}|0\rangle + |1\rangle)$. Find the probabilities of ± 1 and the expectation $\langle \sigma_z \rangle$.

Exercise 10.21 (computational) Measure $A = \text{diag}(1, 2, 3)$ on $|\psi\rangle = \frac{1}{\sqrt{6}}(1, 1, 2)$. Find the probabilities of the three outcomes and $\langle A \rangle$.

Exercise 10.22 (computational) On $\mathbb{C}^3$ let $A = \text{diag}(2, 2, 5)$. Measure A on $|\psi\rangle = \frac{1}{3}(2, 1, 2)$: find the probabilities of each outcome, $\langle A \rangle$, and the post-measurement state when the outcome is 2.

Exercise 10.23 (computational) Verify that $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ commute, and simultaneously diagonalize them in one orthonormal basis.

Exercise 10.24 (★) Suppose T is normal with $T^3 = T$. Show that the eigenvalues of T lie in $\{0, 1, -1\}$ and that T is self-adjoint.

Exercise 10.25 (proof) For an observable $A = \sum_k \lambda_k P_k$ with the spectral projections P_k , prove that the Born-rule probabilities $\|P_k|\psi\rangle\|^2$ of a normalized state $|\psi\rangle$ sum to 1.